Method	Miscoverage	RMSE	Avg. interval length
Radius-based	0.1379	940.77	2540.26
Posterior-based	0.1379	908.40	3157.48
MCMC baseline	0.0345	947.96	3526.89
Target ($\alpha = 0.1$)	0.1000	—	—

Table 2: Empirical miscoverage, RMSE between the median predicted weekly cases and the observed measles outbreak, and average pointwise interval length for the radius-based parameter set, posterior-based parameter set, and MCMC baseline. The target miscoverage level is $\alpha = 0.1$.

given in Section S3.2 of the Supplementary Material.

5.3 Results

Table 2 reports empirical miscoverage, RMSE between the median predicted trajectory and the observed weekly cases, and average pointwise interval length for the two proposed constructions and the MCMC baseline. Figure 6 displays the proposed bands overlaid on the Malawi data, with miscovered weeks highlighted in orange; the KOH–MCMC band was shown in Figure 1a.

Both proposed parameter sets achieve an empirical miscoverage of 0.1379, moderately above the target $\alpha = 0.1$ and consistent with the CRC guarantee $\mathbb{E}[L_{n+1}(\hat{\lambda})] \leq \alpha$, with average interval lengths of 2540.26 and 3157.48 cases. The MCMC baseline attains a miscoverage of 0.0345, less than half the nominal level, with the widest band (3526.89 cases); this pattern is consistent with the sensitivity of posterior uncertainty to the assumed discrepancy specification (Brynjarsdóttir and O'Hagan, 2014, Syring and Martin, 2019). The CRC procedure instead calibrates $\hat{\lambda}$ directly against the empirical miscoverage of synthetic data generated under the heterogeneous GP family, sidestepping the need for any single parametric noise model to be exactly correct. The RMSE values of the three methods are comparable. The ablation study in Section S4 of the Supplementary Material isolates the role of the CRC step on these data: retaining a fixed 90% of the generated draws without CRC, under either selection rule, attains zero miscoverage on the evaluated weeks but produces bands considerably wider than both the CRC and KOH–MCMC bands, indicating overcoverage (cf. Figure 1b).

A further feature of the framework is that the miscovered weeks, shown as orange bands in Figure 6, localize where the SEIR model struggles most. In both CRC panels the orange bands